

T014 Validation Report

Task: T014 — Migration: Meter, SIMCard, MeterAssignment, SIMAssignment in backend/prisma/

Date: 2026-05-27

Branch: feature/t014-meter-sim-migration

Verdict: PASS

Implementation Summary

#	Component	Table/Type	Status
1	Meter model	meters	CREATED
2	SIMCard model	sim_cards	CREATED
3	MeterAssignment model	meter_assignments	CREATED
4	SIMAssignment model	sim_assignments	CREATED
5	MeterType enum	meter_type	CREATED
6	MeterStatus enum	meter_status	CREATED
7	IpType enum	ip_type	CREATED
8	SimStatus enum	sim_status	CREATED
9	AssignmentStatus enum	assignment_status	CREATED

Enums

Enum	Values
meter_type	electricity, water_main, water_child
meter_status	available, assigned, active, offline, faulty, replaced, terminated, retired
ip_type	static, dynamic
sim_status	available, assigned, active, suspended, old, reusable, retired
assignment_status	active, ended

Partial Unique Indexes (FR-004 / FR-005)

Index	Table	Column	Condition
meter_assignments_meter_id_active_key	meter_assignments	meter_id	WHERE end_at IS NULL
sim_assignments_sim_id_active_key	sim_assignments	sim_id	WHERE end_at IS NULL

Foreign Key Constraints

Constraint	From	To	On Delete
meters_parent_main_meter_id_fkey	meters.parent_main_meter_id	meters.id	SET NULL
meter_assignments_meter_id_fkey	meter_assignments.meter_id	meters.id	RESTRICT
sim_assignments_sim_id_fkey	sim_assignments.sim_id	sim_cards.id	RESTRICT
sim_assignments_meter_id_fkey	sim_assignments.meter_id	meters.id	RESTRICT

Validation Results

#	Step	Result	Details
1	npx prisma format	PASS	Schema formatted
2	npx prisma validate	PASS	Schema valid
3	npx prisma migrate dev	PASS	Migration: 20260527100316_meter_sim
4	npx prisma migrate status	PASS	Database up-to-date
5	npm test	PASS	69/69 tests (8 suites)
6	npm run build (tsc)	PASS	Exit code 0
7	Partial unique meter_assignments	CONFIRMED	WHERE end_at IS NULL
8	Partial unique sim_assignments	CONFIRMED	WHERE end_at IS NULL
9	Unique serial_number	CONFIRMED	meters_serial_number_key
10	Unique iccid	CONFIRMED	sim_cards_iccid_key

Acceptance Criteria

#	Criterion	Status
1	Unique serial_number on meters	PASS
2	Unique iccid on sim_cards	PASS
3	Partial unique (meter_id) WHERE end_at IS NULL	PASS (FR-004)
4	Partial unique (sim_id) WHERE end_at IS NULL	PASS (FR-005)
5	parent_main_meter_id nullable, self-FK	PASS
6	All 5 enums created	PASS
7	All 4 tables created	PASS
8	FK constraints within T014	PASS
9	Build stable	PASS
10	Tests passing	PASS

OVERALL: PASS